

CENG 222

Statistical Methods for Computer Engineering

Spring 2024-2025

Homework IV

Due date: 04/06/2025, Wednesday, 23:59

Introduction

In this homework, there are two questions related to Monte Carlo simulations. The first is a classical question that requires written calculations, while the second is a programming question. For the first question, **show your work** and the steps of your calculations. Give an explanation about what numbers mean in those steps. Also, include the code for the programming question in your report as appendix.

Questions

Cafe CENG is operating on campus and classifies its customers in three categories; students, instructor and entrepreneurs. The number of customers attending the cafe follows Poisson distribution for each category. Lambda values for these distributions are;

$\lambda_S = 40$, $\lambda_I = 15$ and $\lambda_E = 5$. S, I and E represents students, instructor and entrepreneurs respectively.

The amount of money a students spends is a Gamma distributed random variable in Liras with $\alpha = 15$ and $\lambda = 0.15$.

The amount of money an instructor spends is a Gamma distributed random variable in Liras with $\alpha = 8$ and $\lambda = 0.05$.

The amount of money an entrepreneur spends follows normal distribution with a mean of 400 Liras and a standart deviation of 50 Liras.

The management of the cafe considers the turnout of the day successful if at least 9000 Liras are spent by the customers. You are asked to estimate the probability of successful turnout by conducting a Monte Carlo simulation study.

Q1. (15 pts.)

(a) Use Normal approximation to determine the size of your Monte Carlo simulation so that with 0.99 probability, your answer should differ from the true value by no more than 0.02. (5pts)

(b) Answer the following questions; (2.5pts each)

- What is the expected value for the money a single student spends?
- What is the expected value for the money a single entrepreneur spends?

- What is the expected value for the total money spent by all instructors on a day?
- What is the expected value for the total money spent by all customers on a day?

Q2. (85 pts.)

Implement a Matlab/Octave code that does this simulation. As the size of the simulation use your answer in part **a** of Q1. In your report provide the following;

- Estimate the probability that the cafe is successful (the total money spent by all customers is more than 9000 Liras).
- Estimate the total money spent by all customers on a day, X .
- Estimate $Std(X)$ and comment on the accuracy of your estimator of X .
- The owners of the cafe came up with a new menu. They expect a 10% increase in the total money spent by students, however they expect a 5% drop in the total money spent by entrepreneurs. How will this expect the chance of success? Briefly comment.
- You are asked to recommend a new success threshold where the success probability is at least 90%. Try to find a new threshold and explain how you found it. The threshold should be as high as possible.

For this question, along the standard report template you are also given a template "hw4.m". You need to fill out necessary parts of this template file. **Provide a screenshot of the output of fprintf lines of the template file (three lines at the bottom) in your report.**

You can use Example 5.9 from the book to sample from the Poisson distribution and Example 5.11 to sample from the Gamma distribution. For normal distribution, you can use *normrnd* function in package *statistics*.

Specifications

- You are expected to write your answers in LaTeX format. You can use the given template.
- Please do not skip the calculation steps. Show every step of your work.
- You have a total of 2 late days for this homework. For each day you have submitted late, you will lose 25 points. If you submit your homework at least 2 days later than the deadline, you will get zero.
- Cheating is forbidden. The violators will be punished according to the department regulations.
- Provide your code as an appendix in your report.
- Follow the course page on ODTUClass for any updates and clarifications. Please ask your questions on ODTUClass instead of e-mailing if they do not contain some part of the solution. If they contain, you can send an email to "mduymus@ceng.metu.edu.tr".

Submission

Submissions will be done via ODTUClass. You are expected to submit a **single** PDF file named "hw4.pdf".